Method of expressing concentration of solution

121. An X molal solution of a compound in benzene has mole fraction of solute equal to 0.2. The value of X is
(a) 14 (b) 3.2
(c) 4 (d) 2

122. Molecular weight of urea is 60. A solution of urea containing $6g$ urea in one litre is
(a) 1 molar (b) 1.5 molar
(c) 0.1 molar (d) 0.01 molar

123. The molar solution of sulphuric acid is equal to
(a) Nsolution (b) $2N$solution
(c) $N/2$ solution (d) $3N$solution

124. The weight of sodium carbonate required to prepare 500 *ml* of a semi- normal solution is
(a) 13.25 *g* (b) 26.5 *g*
(c) 53 *g* (d) 6.125 *g*

125. $200ml$ of a solution contains $5.85g$ dissolved sodium chloride. The concentration of the solution will be
(a) 1 molar (b) 2 molar
(c) 0.5 molar (d) 0.25 molar

126. Molarity of a solution prepared by dissolving 75.5 *g* of pure *KOH* in 540 *ml* solution is
(a) 3.05 *M*
(b) 1.35 *M*
(c) 2.50 *M*
(d) 4.50 *M*

127. Which one of the following is an extensive property
(a) Molar volume
(b) Molarity
(c) Number of moles
(d) Mole fraction

128. Addition of conc. *HCl* to saturated $BaCl_2$ solution precipitates $BaCl_2$; because
(a) It follows from Le Chatelier's principle
(b) Of common-ion effect
(c) Ionic product $(Ba^{++}), (Cl^{-})$ remains constant in a saturated solution
(d) At constant temperature, the product (Ba^{2+}), $(Cl^{-})^2$ remains constant in a saturated solution

129. How much water is needed to dilute 10 *ml* of 10 *N* hydrochloric acid to make it exactly decinormal (0.1 *N*)
(a) 990 ml (b) 1000 ml
(c) 1010 ml (d) 100 ml

130. The formula weight of H_2SO_4 is 98. The weight of the acid in $400ml$ of $0.1M$ solution is
(a) $2.45g$ (b) $3.92g$
(c) $4.90g$ (d) $9.8g$

131. The molarity of pure water is
(a) 55.6 (b) 5.56
(c) 100 (d) 18

132. The molarity of a $0.2NNa_2CO_3$ solution will be
(a) 0.05 M (b) 0.2 M
(c) 0.1 M (d) 0.4 M

133. How many moles of water are present in 180 g of water
(a) 1 mole (b) 18 mole
(c) 10 mole (d) 100 mole

134. If we take $44g$ of CO_2 and $14g$ of N_2 what will be mole fraction of CO_2 in the mixture
(a) 1/5 (b) 1/3
(c) 2/3 (d) 1/4

135. What is the volume of $0.1NHCl$ required to react completely with $1.0g$ of pure calcium carbonate ($Ca = 40, C = 12$ and $O = 16$)
(a) $150cm^3$ (b) $250cm^3$
(c) $200cm^3$ (d) $100cm^3$

136. The amount of $NaOH$ in gms in $250cm^3$ of a $0.100MNaOH$ solution would be
(a) 4 *gm* (b) 2 *gm*
(c) 1 *gm* (d) 2.5 *gm*

137. 4.0 *gm* of $NaOH$ are contained in one decilitre of solution. Its molarity would be
(a) 4 *M* (b) 2 *M*
(c) 1 *M* (d) 1.5 *M*

138. When 90 *gm* of water is mixed with 300 *gm* of acetic acid. The total number of moles will be
(a) 5 (b) 10
(c) 15 (d) 20

139. A molal solution is one that contains one mole of a solute in
(a) 1000 gm of the solvent
(b) One litre of the solvent
(c) One litre of the solution
(d) 22.4 litres of the solution

140. What weight of ferrous ammonium sulphate is needed to prepare 100 *ml* of 0.1 normal solution (mol. wt. 392)
(a) 39.2 gm (b) 3.92 gm
(c) 1.96 gm (d) 19.6 gm

141. If 18 gm of glucose $(C_6H_{12}O_6)$ is present in 1000 *gm* of an aqueous solution of glucose, it is said to be
(a) 1 molal (b) 1.1 molal
(c) 0.5 molal (d) 0.1 molal

142. The number of moles of KCl in $1000ml$ of 3 molar solution is
(a) 1 (b) 2
(c) 3 (d) 1.5

143. The unit of molality is
(a) Mole per litre
(b) Mole per kilogram
(c) Per mole per litre
(d) Mole litre

144. A solution contains 1 mole of water and 4 mole of ethanol. The mole fraction of water and ethanol will be
(a) 0.2 water + 0.8 ethanol
(b) 0.4 water + 0.6 ethanol
(c) 0.6 water + 0.8 ethanol
(d) 0.8 water + 0.2 ethanol